

## Mortgage Loan Management System (MLMS)

### End-to-End Azure Cloud Microservices Case Study

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#### 1. Detailed User Stories (Expanded & Simple)

User stories are written using **Agile format** and grouped by **role**.

They are **intentionally limited in complexity** and suitable for a **5-day execution window**.

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#### 2. Customer User Stories

##### US-C1: Customer Registration

###### As a Customer

I want to register with my email and password

So that I can log in and apply for a mortgage loan.

###### Acceptance Criteria

- User is stored in Cosmos DB
  - Default role = Customer
  - Password is stored as hash
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##### US-C2: Customer Login

###### As a Customer

I want to log in to the application

So that I can securely access my dashboard.

###### Acceptance Criteria

- JWT token generated
  - Token contains user role
  - Token expiry configured
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##### US-C3: Create Customer Profile

**As a Customer**

I want to create my basic profile

So that the bank has my personal information.

**Acceptance Criteria**

- Name, Email, Phone saved in Azure SQL
  - Customer linked to logged-in user
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**US-C4: Apply for Mortgage Loan****As a Customer**

I want to apply for a mortgage loan

So that the bank can review my request.

**Acceptance Criteria**

- Loan amount and property address saved
  - Initial status = Submitted
  - LoanSubmitted event published to Service Bus
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**US-C5: View Loan Status****As a Customer**

I want to view the status of my loan

So that I know whether it is approved or rejected.

**Acceptance Criteria**

- Only own loans visible
  - Status shown as Submitted / Approved / Rejected
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**US-C6: Upload Property Document (Blob)****As a Customer**

I want to upload a property document

So that the loan officer can verify details.

## **Acceptance Criteria**

- File stored in Blob Storage
  - Blob URL stored against LoanId
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## **3. Loan Officer User Stories**

### **US-L1: View All Loan Applications**

#### **As a Loan Officer**

I want to view all submitted loan applications  
So that I can review them.

#### **Acceptance Criteria**

- Only Loan Officers can access
  - Loans sorted by submission date
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### **US-L2: Approve Loan**

#### **As a Loan Officer**

I want to approve a loan  
So that the customer can proceed further.

#### **Acceptance Criteria**

- Loan status updated to Approved
  - LoanApproved event published to Service Bus
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### **US-L3: Reject Loan**

#### **As a Loan Officer**

I want to reject a loan  
So that invalid applications are closed.

#### **Acceptance Criteria**

- Loan status updated to Rejected
- Reason optional

- LoanRejected event published
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#### **4. Admin User Stories**

##### **US-A1: Manage Users**

###### **As an Admin**

I want to view all users

So that I can manage access.

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##### **US-A2: Assign Roles**

###### **As an Admin**

I want to assign roles to users

So that they can act as Customer or Loan Officer.

###### **Acceptance Criteria**

- Roles updated in Cosmos DB
- Effective on next login